# "INTRODUCTION TO ENERGY" WORKSHEET

Name_____

Date _____ Block_____

**Part 1. The two basic types of energy**

Directions: Determine the best match between basic types of energy and the description provided. Put the correct letter in the blank.

_____1. A skier at the top of the mountain (a) Kinetic Energy

_____2. Gasoline in a storage tank (b) Potential Energy

_____3. A race-care traveling at its maximum speed (c) Both forms of Energy

_____4. Water flowing from a waterfall before it hits the pond below

_____5. A spring in a pinball machine before it is released

_____6. Burning a match

_____7. A running refrigerator motor

**Part 2. Definitions of Energy.**

Directions: Write down the definition for each of the following terms after reading the article.

ENERGY:

KINETIC ENERGY:

POTENTIAL ENERGY:

**Part 3. Forms of Energy.**

Directions: Determine the type of energy for each form (Kinetic, Potential, or Both) and give an example.

| Form | Definition | Type (KE, PE, or Both) | Example (for each type if both) |
|---|---|---|---|
| Mechanical (motion) energy | An object's movement creates energy | | |
| Thermal (heat) energy | The vibration and movement of molecules | | |
| Radiant energy | Electromagnetic waves | | |
| Electrical energy | Movement of electrons | | |

| | | | |
|---|---|---|---|
| Chemical energy | Stored in bonds of atoms and molecules | | |
| Nuclear energy | Stored in the nucleus of an atom; released when nucleus splits or combines | | |
| Sound energy | Vibration of waves through material | | |
| Gravitational energy | Energy of position or height | | |

**Part 4. Forms of Energy Continued**

Directions: Match the energy form(s) to the description provided. A few questions may have more than one answer.

_____1. Falling rocks from the top of a mountain

_____2. Release of energy from the Sun

_____3. Energy released from food after it is eaten

_____4. Batteries

_____5. The energy that runs a refrigerator

_____6. Nuclear fission reactors

_____7. The rumble of thunder from a storm

_____8. Rubbing your hands together

_____9. Gasoline

_____10. Food before it is eaten

_____11. Lightening

(a) Mechanical

(b) Electrical

(c) Heat

(d) Radiant

(e) Chemical

(f) Nuclear

(g) Sound

**Part 5. Transformation of Energy**

Directions: Use the following forms of energy to fill in the table below: **mechanical, electrical, heat, radiant, chemical, nuclear, and sound**. The first one has been done for you.

| | **ORIGINAL ENERGY FORM** | **FINAL ENERGY FORM** |
|---|---|---|
| 1. Electric motor | electrical | mechanical |
| 2. A battery that runs a moving toy | | |
| 3. A solar panel on the roof of a house | | |

| | | | |
|---|---|---|---|
| 4. | A person lifting a chair | | |
| 5. | A nuclear power plant | | |
| 6. | A toaster | | |
| 7. | A church bell | | |
| 8. | Gasoline powering a car | | |
| 9. | A light bulb | | |
| 10. | Photosynthesis | | |